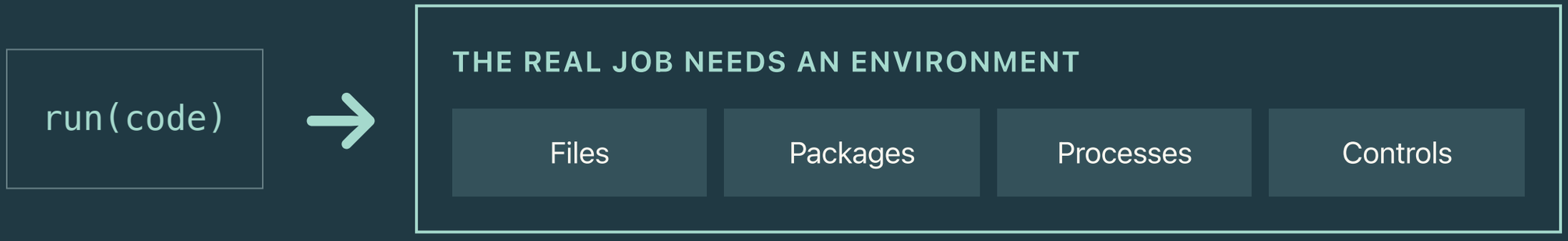
Your agent ran the code. Did it finish the job?

Correct outputs. Useful features. Data boundaries.
A practical comparison of agent execution environments.



Repeated pilot + documented product capabilities · Reviewed 13 October 2026

“Run this code” becomes “finish this workflow.”

An analyst’s export can turn into a coding agent’s workspace.

01

Read

Upload files.
Inspect the data.

02

Transform

Install packages.
Clean and query.

03

Build

Edit code.
Run tests and servers.

04

Continue

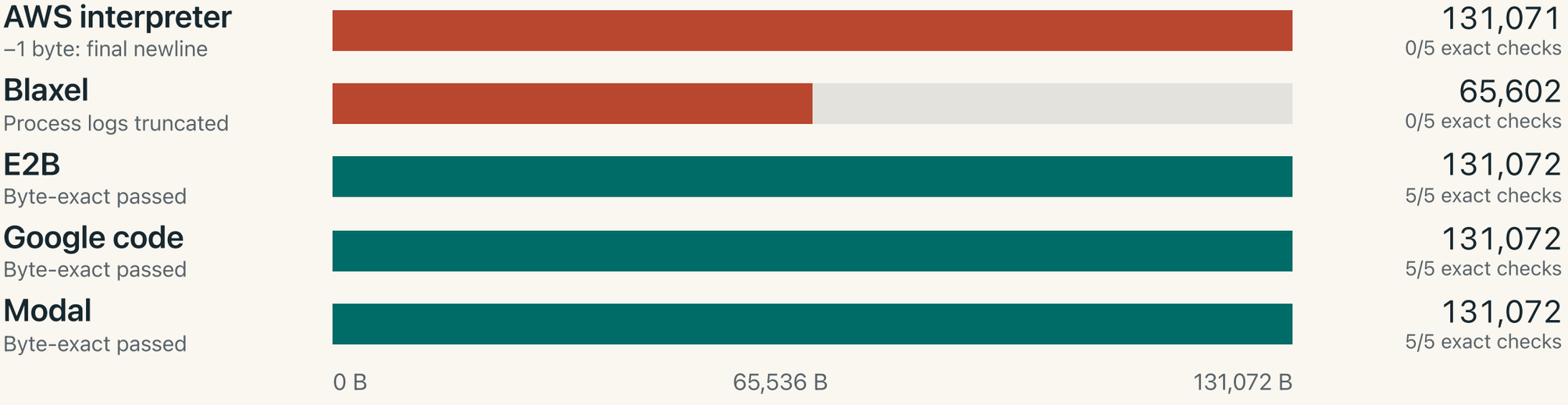
Return later.
Recover the state.

> Each step adds a requirement. **Choose for the whole job.**

Illustrative workflow. The pilot tested component fixtures, not this complete application.

A successful response can still change the result.

The fixture printed exactly **`131,072 bytes`** of stdout.



One missing byte can fail exactness. Truncated logs are a different failure.

Five trials per setup. These are tested output paths—not provider-wide file limits. Full bytes alone do not prove correctness.

Full evidence + source notes: companion HTML / source-guide package.

Three integrations passed three useful workflow fixtures.

A reason to investigate—not an overall winner.

Integration	Binary files	10k-row ETL	Offline install
E2B	**5/5** passed / evaluated	**5/5** passed / evaluated	**5/5** passed / evaluated
Modal	**5/5** passed / evaluated	**5/5** passed / evaluated	**5/5** passed / evaluated
Google code	**5/5** passed / evaluated	**5/5** passed / evaluated	**5/5** passed / evaluated

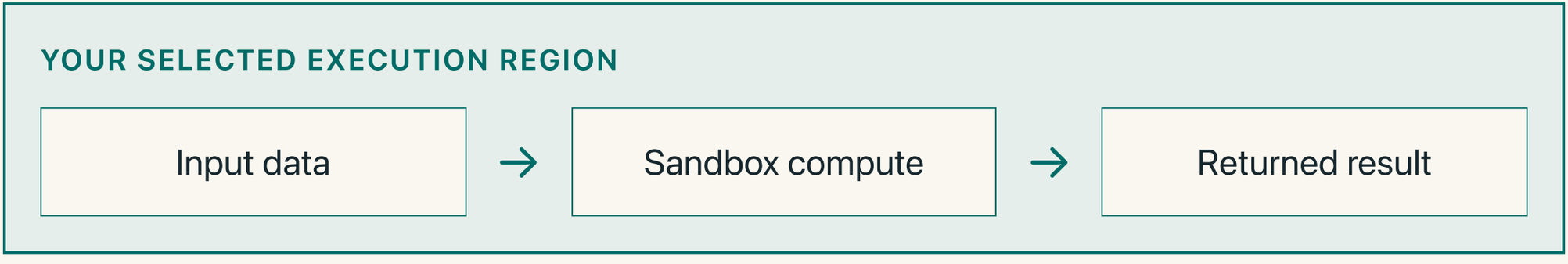
> E2B and Modal also passed **cross-call file state and recovery** fixtures.

Five-trial fixtures, not production reliability. Installation: one offline pure-Python wheel. State: files, not restored memory.

Full evidence + source notes: companion HTML / source-guide package.

EU compute is only one part of EU data residency.

A working integration still needs an eligible deployment.



Stored copies

Files, images, snapshots, backups

Operations

Logs, metadata, support access

External calls

Model inference, APIs, downloads

> A region selector does not define **every data boundary.**

Conceptual data-flow map. Residency commitments must identify data categories, processing locations and access scope.

Full evidence + source notes: companion HTML / source-guide package.

European placement is available. The boundaries differ.

Product	Documented EU / European route	Boundary to investigate
Modal	EEA compute pinning	US logs, snapshots + storage.
E2B	Belgium · Pro+	Support-enabled; wider scope unknown.
Blaxel	Frankfurt · Mark 3	Volumes regional; wider scope unknown.
Google	EU regions / eu endpoint	Model + keys: separate configuration.
AWS	EU Built-in Tools	Check endpoint, storage + retention.
Databricks	EU workspace regions	Cloud + execution surface matter.
Daytona	Shared eu target	Shared GPU not EU-pinnable.

Reviewed 13 October 2026. Availability is product-, plan- and class-specific. EEA / “Europe” labels are not strict EU-member-state guarantees.

Full evidence + source notes: companion HTML / source-guide package.

This was not an EU-only benchmark.

Available regions cannot fill gaps in recorded deployment evidence.

AWS	Requested	Stockholm · `eu-north-1`
Blaxel	Requested	Oregon · `us-pdx-1`
Modal	Unpinned	Resolved execution region unknown.
Google	Unretained	Campaign refers to a missing region preflight.
E2B / Databricks SQL	Unknown	Cluster / warehouse region not retained.

Requested location is not independent placement attestation. Daytona and dedicated Databricks lacked evaluated remote baselines.

Full evidence + source notes: companion HTML / source-guide package.

Who owns the environment changes the trade-off.

MODEL-OWNED

Execute inside the answer.

Gemini / OpenAI tools

Gain: less orchestration.
Give up: OS and lifecycle control.

MANAGED

Your app owns the calls.

Workspace APIs / cloud interpreters

Gain: direct execution.
Give up: some deployment control.

SELF-OPERATED

Your team owns the fleet.

Kubernetes / OpenSandbox / VMs

Gain: infrastructure policy.
Take on: capacity and operations.

> Compare the **exact execution product**—not the cloud logo.

Custom containers and customer compute are different BYOC meanings. A vendor-managed control plane may remain.

Modal: configure the workspace around the workload.

Custom toolchains, resource knobs and an explicit GPU option.

WHAT YOU GAIN

Custom images. Independent CPU / RAM controls. Volumes and authenticated previews.

LIMITS AND TRADE-OFFS

Sandbox GPUs are preemptible and require gVisor. Advanced paths were not tested.

Pilot Files and offline install passed. GPU / durable state: documented only.

EU BOUNDARY · DOCUMENTED, NOT ATTESTED

EEA compute; US logs, snapshots and managed storage. Lifecycle control plane: US-east.

Region / residency source

Capabilities are documented unless marked measured. Conditions are in the official sources and companion guide.

Full evidence + source notes: companion HTML / source-guide package.

E2B: a workspace for iterative CPU work.

Files, Python dependencies and repeated calls into one workspace.

WHAT YOU GAIN

Image-derived templates. Preview URLs. Documented memory + file pause/resume.

LIMITS AND TRADE-OFFS

CPU-only offering. Runtime limits depend on plan; pause/resume was not tested.

Pilot Files, offline install, shared disk state and recovery passed. One deadline response was late.

EU BOUNDARY · DOCUMENTED, NOT ATTESTED

Shared Belgium cluster: Pro+ via support. Enterprise BYOC documents customer-VPC storage.

Region / residency source

Capabilities are documented unless marked measured. Conditions are in the official sources and companion guide.

Full evidence + source notes: companion HTML / source-guide package.

Blaxel: build around lifecycle and previews.

Standby workspaces, persistent volumes and application URLs.

WHAT YOU GAIN

Standby / restore. Named processes. Public and private previews.

LIMITS AND TRADE-OFFS

CPU follows memory sizing. Custom images require sandbox-api.

Pilot This adapter returned truncated process logs and uses lossy text-based binary staging.

EU BOUNDARY · DOCUMENTED, NOT ATTESTED

Frankfurt is documented for Mark 3. Volumes are regional; wider data scope is unverified.

Region / residency source

Capabilities are documented unless marked measured. Conditions are in the official sources and companion guide.

Full evidence + source notes: companion HTML / source-guide package.

Google: default code and BYOC are different paths.

Bounded Python or a custom container? Choose the API path.

DEFAULT CODE

Bounded Python; no network. API machine configurations are documented.

BYOC PATH

Custom images and environment controls. Separate deployment; untested here.

Pilot Default-code artifacts / data passed. This adapter did not request a machine configuration.

EU BOUNDARY · DOCUMENTED, NOT ATTESTED

EU regions / eu endpoint; model location separate. CMEK needs the documented single-region setup.

Region / residency source Code API sizing CMEK conditions

Capabilities are documented unless marked measured. Conditions are in the official sources and companion guide.

Full evidence + source notes: companion HTML / source-guide package.

AWS: a direct interpreter with cloud connectivity.

Managed execution with AWS identity, networking and storage.

WHAT YOU GAIN

Direct SDK calls. Documented VPC networking and customer-owned S3 / EFS mounts.

LIMITS AND TRADE-OFFS

No custom image or caller CPU / RAM sizing in the reviewed interpreter APIs.

Pilot Code-edit and SQLite passed. Stdout lost a final newline in 5/5 trials.

EU BOUNDARY · DOCUMENTED, NOT ATTESTED

Pilot requested Stockholm. Session docs list 30-day retention; stop is not deletion attestation.

Region / residency source Session retention

Capabilities are documented unless marked measured. Conditions are in the official sources and companion guide.

Full evidence + source notes: companion HTML / source-guide package.

Databricks: choose the execution surface first.

Three routes. Different state, networking and evidence.

Documented Beta

Dedicated Sandbox

4 vCPU / 8 GB. Home persists. Unrestricted egress.

Measured here

SQL python_exec

Stateless Python through a warehouse. No artifact collector.

Documented Beta

Sandbox MCP

Python / SQL / shell. No network egress.

Scope SQL python_exec was measured. Dedicated API unavailable; sandbox MCP untested.

EU BOUNDARY · DOCUMENTED, NOT ATTESTED

Supported regions vary by cloud / surface. AI Europe Geo includes UK + Switzerland; SQL region unrecorded.

Region / residency source Execution surfaces

Capabilities are documented unless marked measured. Conditions are in the official sources and companion guide.

Full evidence + source notes: companion HTML / source-guide package.

Daytona: the sandbox class changes the contract.

Development workspaces with images, previews or a GPU class.

WHAT YOU GAIN

OCI images, sizing and signed previews. Container disk / VM memory have distinct lifecycles.

LIMITS AND TRADE-OFFS

GPU stop deletes the sandbox. Shared GPU placement ignores region preference.

Scope Documented candidate. No evaluated remote baseline; client initialization failed.

EU BOUNDARY · DOCUMENTED, NOT ATTESTED

Shared eu target exists. EU-pinned shared GPUs are not offered by this placement contract.

Region / residency source

Capabilities are documented unless marked measured. Conditions are in the official sources and companion guide.

Full evidence + source notes: companion HTML / source-guide package.

Keeping files is not the same as restoring memory.

Three separate contracts. Ask what survives each lifecycle event.

Contract	What it preserves	What the pilot proves
Live files	**Files between calls**	E2B / Modal: tested.
Durable disk	**Files after stop/start**	Documented options. Not tested.
Memory resume	**Running state**	E2B / Daytona VM docs. Not tested.

> Daytona’s GPU class is **deleted on stop**. Class changes the contract.

Retained disk, snapshots and volumes can remain billable without active compute. Session reuse does not prove Python globals persist.

Full evidence + source notes: companion HTML / source-guide package.

Less orchestration—or more deployment ownership?

GEMINI / OPENAI TOOLS

Let the model run code.

Pro: less orchestration.
Con: fixed environment.

OpenAI: EEA + Switzerland, eligibility required; system data excluded.
Gemini Developer API: no EU-only boundary established.

KUBERNETES / OPENSANDBOX

Operate your own fleet.

Pro: choose infrastructure policy.
Con: operate the fleet.

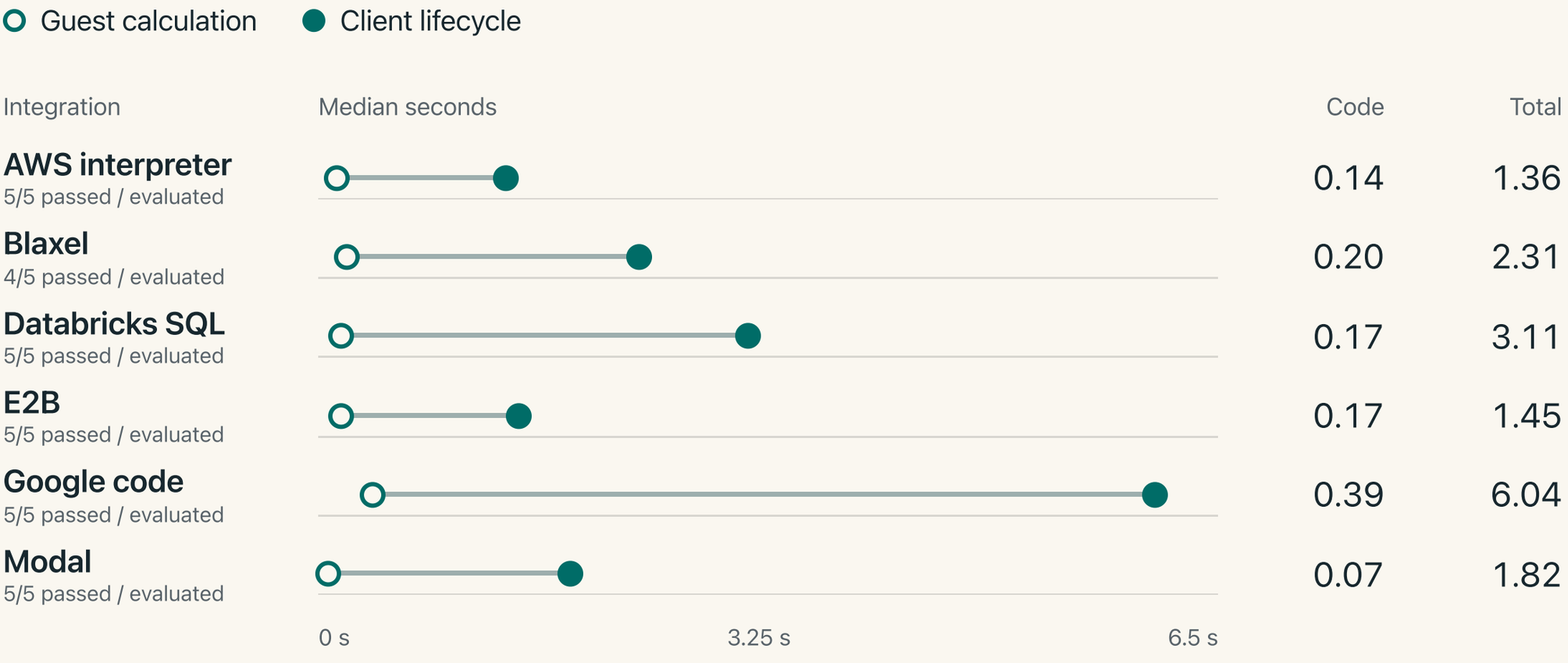
EU-hosted compute still needs EU-scoped logs, support and downstream calls.

Current OpenAI Responses containers were not benchmarked. No attested Kubernetes / remote OpenSandbox deployment comparison.

OpenAI data controls Gemini API terms

Full evidence + source notes: companion HTML / source-guide package.

Fast calculation is not fast completion.



Successful-trial medians; same fixture, different setups / regions. The gap is not measured overhead. Five-trial pilot: no tail-latency claim or equal-hardware ranking.

Full evidence + source notes: companion HTML / source-guide package.

A deadline response does not prove remote cancellation.

A 4-second request. A 9-second response budget.

MODAL

4.21–4.22 s

5/5 passed the response oracle.

E2B

8.78–10.88 s

4/5 passed the response oracle.

> Check **when the caller hears back** and **whether the work stops**.

Execute-phase response ranges, five trials per setup. This oracle did not independently attest remote process termination.

Full evidence + source notes: companion HTML / source-guide package.

A cheap session is not a cheap completed task.

YOUR TOTAL CAMPAIGN BILL

- Active resources +
- Idle / retained storage +
- Retries and failures +
- Model usage / fleet ops +

Total cost

÷

Verified completions

Unmeasured

No invoice-backed task-cost ranking is available.

List rates have incompatible units: physical cores, vCPUs, RAM, session windows and tokens. Keep promotional credits separate from gross usage.

Full evidence + source notes: companion HTML / source-guide package.

GDPR is not a blanket EU-only hosting rule.

EU-only may be your contractual or sector-specific requirement.
International transfers need their own valid basis.

1 **Define the geographic boundary.**

EU members, EEA or wider Europe? Include storage and processing.

2 **Check processor and transfer terms.**

DPA, subprocessors, remote access, adequacy / safeguards where applicable.

3 **Verify the configured data flows.**

Compute, logs, backups, retention, deletion and downstream APIs.

GDPR Articles 28, 32, 44–46. EU compute, encryption or BYOC alone does not establish compliance. The pilot did not attest these controls.

GDPR EDPB transfer guide

Product fit earns a shortlist. EU evidence gates deployment.

CPU iteration	E2B / Modal · files, packages and repeated calls.
Images / GPUs	Modal / Daytona · verify class, placement and capacity.
Standby / previews	Blaxel · validate binary and output transport.
Cloud connectivity	Google / AWS · select the exact execution path.
Databricks workflows	Choose SQL, dedicated Sandbox or sandbox MCP first.
Fleet ownership	Kubernetes / OpenSandbox / VMs · include operator cost.

Conditional investigation routes, not rankings or EU-compliance endorsements. Model-owned tools are another route for short code-assisted answers.

Choose the smallest environment that can finish your real job.

One workflow. Two eligible candidates.
Pinned deployments. Verified results.

WHAT WILL CHANGE YOUR SHORTLIST?

- EU boundary
- State
- GPU
- Previews

Test your files, recovery and cleanup.
Measure cost per correct completion.

975 attempts retained · 555 carry evaluator hashes · unequal task coverage, not a reliability ranking.

Full evidence + source notes: companion HTML / source-guide package.